

Note: This handout has exercises but those are not to be submitted. They are intended to be instructional and clarify the construction of natural numbers.

ON THE BASIC ASSUMPTIONS ABOUT NATURAL NUMBERS

This section is meant to be a review of what we are willing to take for granted about natural numbers. In mathematics the following structure is paramount: we assume some basic facts about our objects of study (“axioms”), and everything else must be derived from these basic facts alone (“theorems”). In below we lay our working assumptions about natural numbers. They are not as carefully laid as should be in a Logic course – but that care is part of MTH303 and not intended to be laid bare here.

We will assume that there exists a **non-empty set**, denoted $\mathbb{N}$ with the following properties:

- ★ There is a binary operator $+$ on $\mathbb{N}$ which is
 - associative $a + (b + c) = (a + b) + c$
 - commutative $a + b = b + a$
 - satisfies cancellation $a + x = b + x \Rightarrow a = b$.
- ★ There is a binary operator $\times$ on $\mathbb{N}$ which is
 - associative $a \times (b \times c) = (a \times b) \times c$
 - commutative $a \times b = b \times a$
 - satisfies cancellation $a \times x = b \times x \Rightarrow a = b$.
- ★ ($+$ and $\times$ are related) • distributive law: $a \times (b + c) = a \times b + a \times c$. We also have $(b + c) \times a = b \times a + c \times a$, but it can be deduced by commutativity.

Now using this we can define a relation $<$ on $\mathbb{N}$: $a < b$ if $\exists x \in \mathbb{N}$ such that $a + x = b$. It is further assumed that:

- ★ ($<$ is well ordered): • For any non-empty $S \subset \mathbb{N}$ there is a minimum element in S w.r.t. $<$ above.
(Here by minimum we mean an $x \in S$ such that for all $y \in S$ with $y \neq x$, we have $x < y$).

It is conventional to denote that minimum of $\mathbb{N}$ as 1. One further assumes following seemingly obvious properties:

- ★ $1 \not< 1$.
- ★ $1 \times n = n \times 1 = n$ for all $n \in \mathbb{N}$.

Exercise 1. Only from above assumptions, show that $<$ is indeed a total order, that is:

- (1) (anti reflexive) $a < a$ is false for any a . (Hint: Reduce it to the case $1 < 1$).
- (2) (anti symmetric) $a < b$, then $b \not< a$.
- (3) (transitive) $a < b$ and $b < c$ then $a < c$.
- (4) (total) For any two $a, b \in \mathbb{N}$, one of $a < b$ or $b < a$ or $a = b$ must hold.

The two obvious properties are necessary to be assumed, as the following exercise shows:

Exercise 2. Show that

- (1) there is a set, different from the usual $\mathbb{N}$, which satisfies every property above except $1 \not< 1$.
- (2) there is a set, different from the usual $\mathbb{N}$, which satisfies every property above except $1 \times n = n \times 1 = n$.

Note. It is conventional to define successor of n as $n + 1$. We will also denote successor of n as n^+ .

Exercise 3. Given the function $m \mapsto m^+ : \mathbb{N} \rightarrow \mathbb{N}$, show the following:

- (1) • $\nexists m$ s.t. $m^+ = 1$ • $n \neq 1$ then $\exists m$ s.t. $m^+ = n$ • $m^+ = n^+ \Rightarrow m = n$ and
• principal of induction: if there is a property P of natural numbers such that (• $P(1)$ is true and • $P(n)$ is true then $P(n^+)$ is true.) then ($P(n)$ is true for all $n \in \mathbb{N}$).
- (2) Addition satisfies: $n + 1 = n^+$ and $n + m^+ = (n + m)^+$.
- (3) Multiplication satisfies: $n \times 1 = n$ and $n \times m^+ = n \times m + n$.

Remark. In the previous exercise the properties 1 is a set of axioms that characterize natural numbers, the so called Peano’s axioms. The properties 2 and 3 then define addition and multiplication respectively. Commutativity, associativity, distributivity, cancellation, etc. can then be deduced from these properties alone and you may want to try this as an exercise (the key is the principal of induction!).

Remark. What about integers? In classroom we actually assumed certain basic properties of integers. However those can be derived from the properties above if we take the following definition:

We define integers $\mathbb{Z}$ as follows:

- First defined $\mathbb{N} \times \mathbb{N}$ to be the set of ordered pairs: $\{(m, n) \mid m, n \in \mathbb{N}\}$.
 - Morally, an element (m, n) will play the role of “ $m - n$ ” even though we are jumping the gun and there is no subtraction yet.
 - To achieve this, we define the equivalence relation $\sim$ via $(m, n) \sim (m', n')$ iff $m + n' = m' + n$ (we only need $+$ to define this, not subtraction!).
- Define $\mathbb{Z}$ to be the set of equivalence classes modulo this relation.
 - Recall that the equivalence class of (m, n) , denoted $\overline{(m, n)}$ is the set $\{(m', n') \mid (m, n) \sim (m', n')\}$.
 - Verify that $(m, n) \sim (m', n')$ being an equivalence relation implies that $\overline{(m, n)} = \overline{(m', n')}$, both pairs now indeed represent “ $m - n$ ” = “ $m' - n'$ ” (even though both sides are undefined: no subtraction yet!).
- Now we can define various operations on $\mathbb{Z}$ using the same on $\mathbb{N}$
 - $\overline{(m, n)} + \overline{(m', n')} = \overline{(m + m', n + n')}$ defines addition and $\overline{(m, n)} \times \overline{(m', n')} = \overline{(mm' + nn', mn' + m'n)}$ defines multiplication.
 - Negation is defined by $-\overline{(m, n)} := \overline{(n, m)}$ while we define $0 := \overline{(m, m)}$.
 - The set $\mathbb{N}$ is considered a subset of $\mathbb{Z}$ via the map $n \mapsto \overline{(n + 1, 1)}$. Define $a < b$ if $\exists x \in \mathbb{N}$ such that $a + x = b$.
 - Define $|a| = a$ OR 0 OR $-a$ accordingly as $a > 0$, $a = 0$ OR $a < 0$, for all $a \in \mathbb{Z}$.
 - Define $\text{sign}(a) = +1$ for $a > 0$ and -1 for $a < 0$. We avoid defining sign of 0 for now.

Using this it is now possible to prove the following properties of $\mathbb{Z}$ which we had assumed in class:

Exercise 4. Show that:

- (1) Show that the inclusion of $\mathbb{N}$ in $\mathbb{Z}$ preserves $+$, $\times$, $<$.
- (2) Operations $+$ and $\times$ are associative and commutative on $\mathbb{Z}$. $\times$ is distributive over $+$.
- (3) 0 is the identity for addition (i.e. $x + 0 = x = 0 + x$ for all x) while 1 is the identity for multiplication (i.e. $x \times 1 = 1 \times x = x$ for all x).
- (4) Operations $+$ satisfies cancellation. $\times$ satisfies cancellation of a nonzero element.
- (5) The rule of signs holds: $(-a) \times b = a \times (-b) = -(a \times b)$ and $(-a) \times (-b) = a \times b$.
- (6) The relation $<$ is a total order.
 - Further, $m < n, m' < n'$ then $m + n < m' + n'$.
 - Also for any $r > 0$, $m < n$ if and only if $mr < nr$, while for $r < 0$, $m < n$ if and only if $mr > nr$.
- (7) For all $a \in \mathbb{Z}$, we have $|a| \geq 0$. For any two $a, b \in \mathbb{Z}$, we have the triangle inequality $|a + b| \leq |a| + |b|$ with equality if and only if $\text{sign}(a) = \text{sign}(b)$ or $a = 0$ or $b = 0$.
- (8) Any non-empty subset $S \subset \mathbb{Z}$ has a minimum (resp. maximum) if and only if it has a lower bound (resp. upper bound).

Remark. By a lower bound (resp. upper bound) we mean an element $x \in \mathbb{Z}$ such that for all $y \in S$, $y \neq x$ then $y < x$ (resp. $y > x$). If additionally, $x \in S$, we say that x is the minimum (resp. maximum) of S .

Exercise 5. Show that the following hold (they are all equivalent to well ordering of $\mathbb{N}$).

- (1) (**Principal of weak induction**) If a property P on $\mathbb{N}$ satisfies that if $P(1)$ is true and for all $n \in \mathbb{N}$, $P(n) \Rightarrow P(n + 1)$. Then $P(n)$ is true for all $n \in \mathbb{N}$.
- (2) (**Principal of strong induction**) If a property P on $\mathbb{N}$ satisfies that if $P(1)$ is true and for all $n \in \mathbb{N}$, $(P(m) \text{ for all } m \leq n) \Rightarrow P(n + 1)$. Then $P(n)$ is true for all $n \in \mathbb{N}$.
- (3) (**Proof by infinite descent**) If a property P on $\mathbb{N}$ satisfies that if there is an n such that $P(n)$ is false, then there is an $m < n$ with $P(m)$ false. Then $P(n)$ is true for all $n \in \mathbb{N}$.

Exercise 6. We say a set $S \subset \mathbb{Z}$ is **finite** if there is an integer n such that for all $m \in S$, $-n \leq m \leq n$. Show that a finite set of integers (and in particular, natural numbers) always has a maximum.